Mechatronics Design Laboratory I

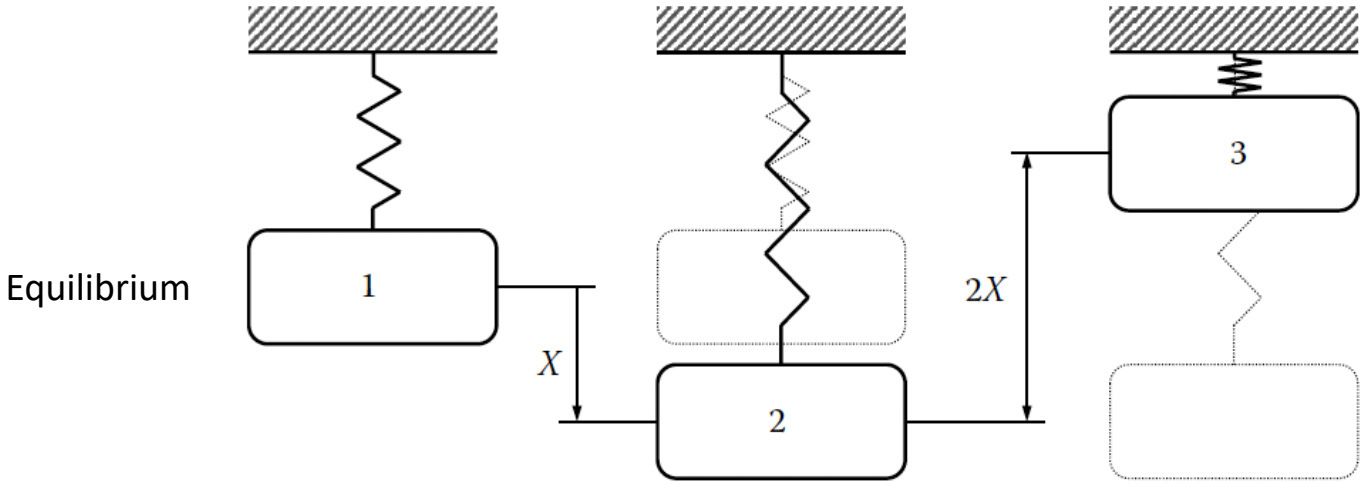
ENME 561/661

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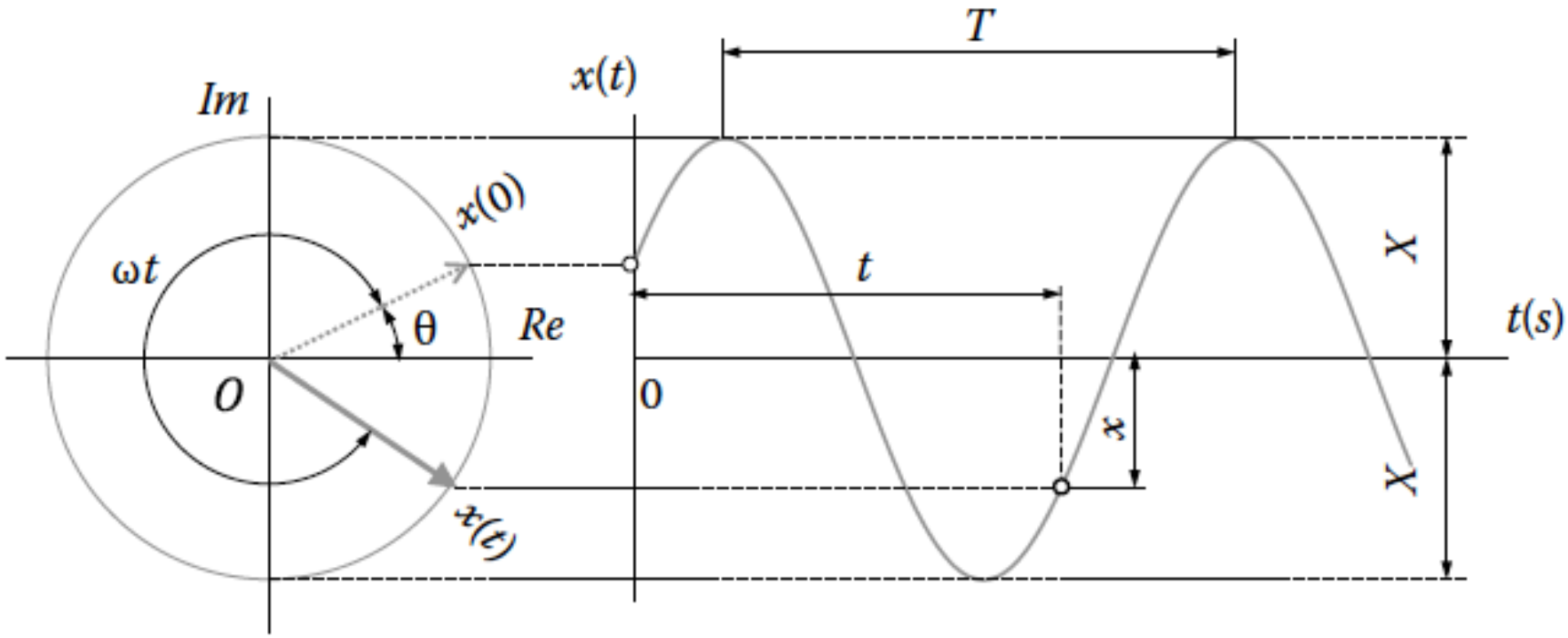
Simple Harmonic



$$x(t) = X sin(\omega t + \theta)$$

Complex Plane

Amplitude versus time plot



$$x(t) = X_{Re}(t) + jX_{Im}(t)$$

$$x(t) = Xsin(\omega t + \theta)$$

$$x_{Re}(t) = Xcos(\omega t + \theta)$$

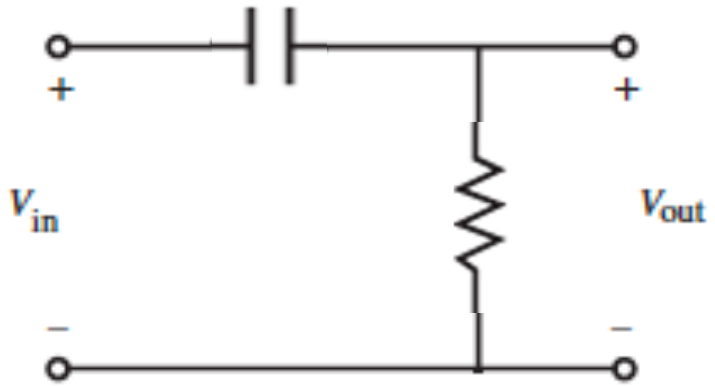
$$x_{Im}(t) = Xsin(\omega t + \theta)$$

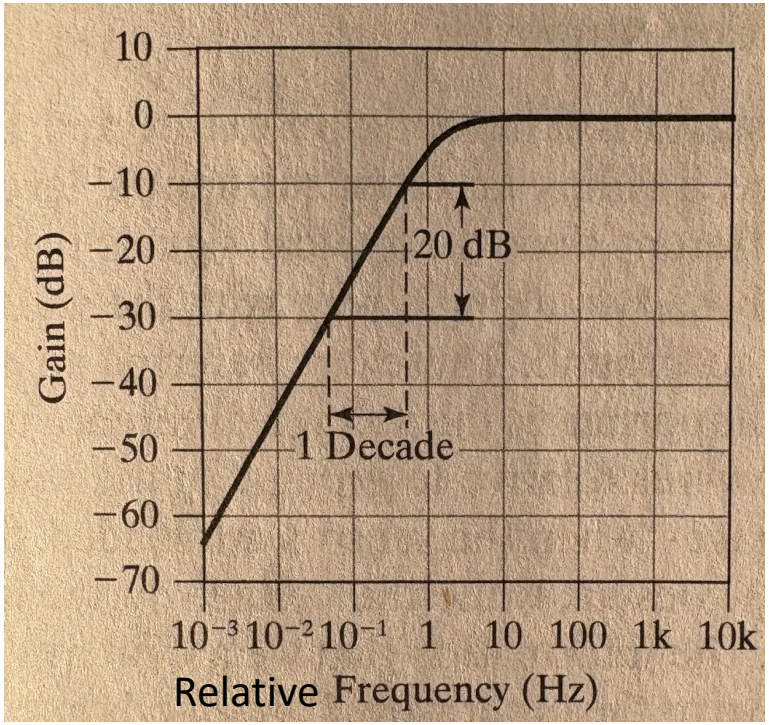
$$X = \sqrt{x_{Re}^2 + x_{Im}^2}$$

$$\theta = tan^{-1}\frac{x_{Im}(t)}{x_{Re}(t)}$$

Analog Passive High Pass Filter

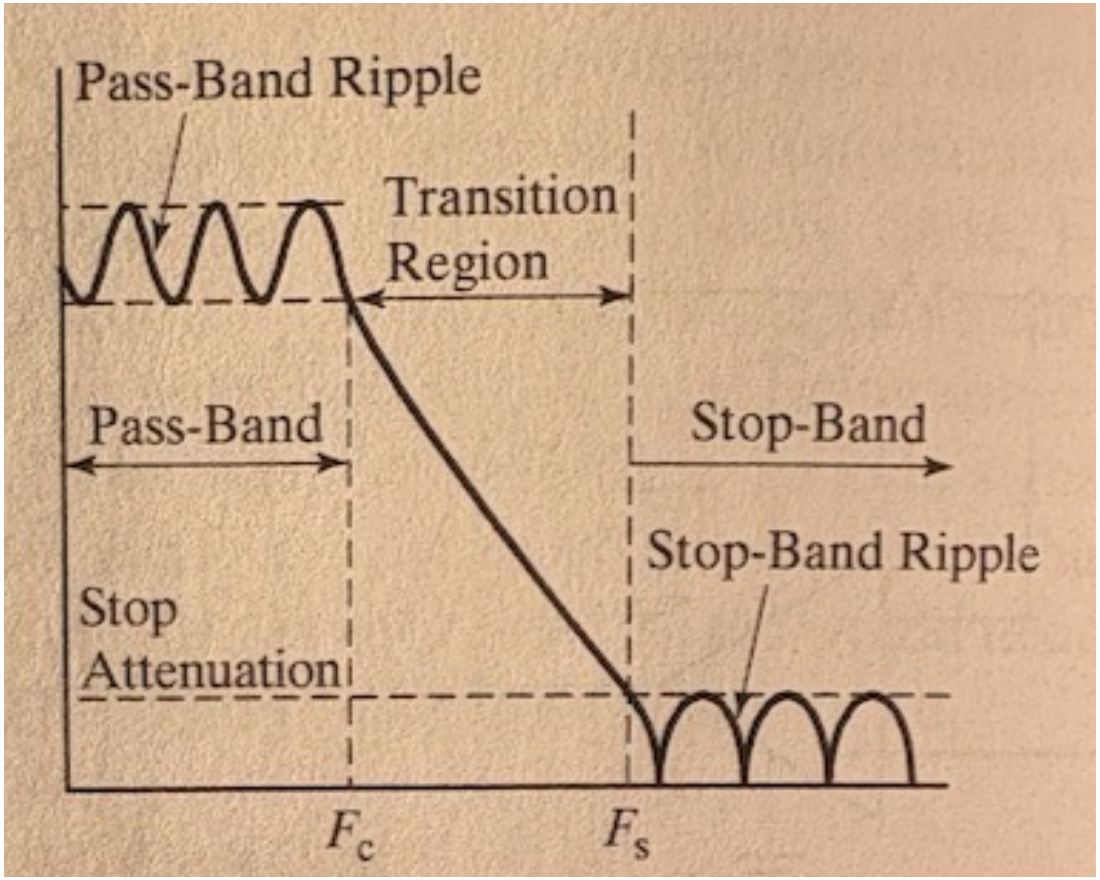
(Example: 1st order high pass RC filter)





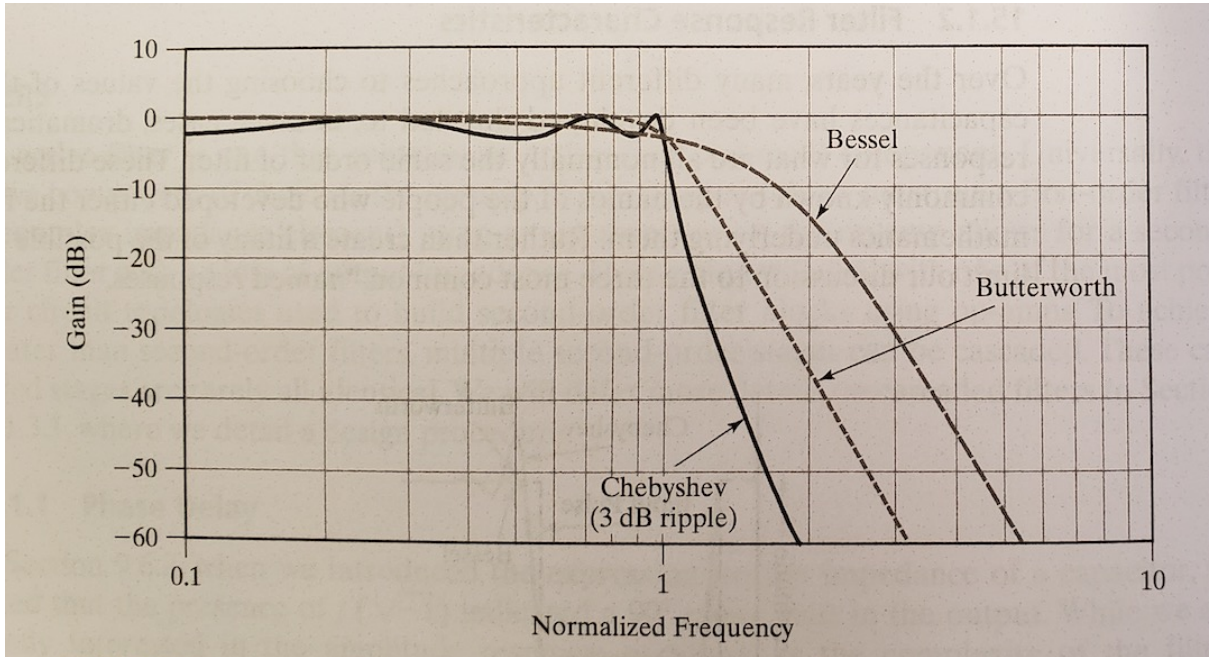
Filter Response Terminology

using a low pass filter as an example

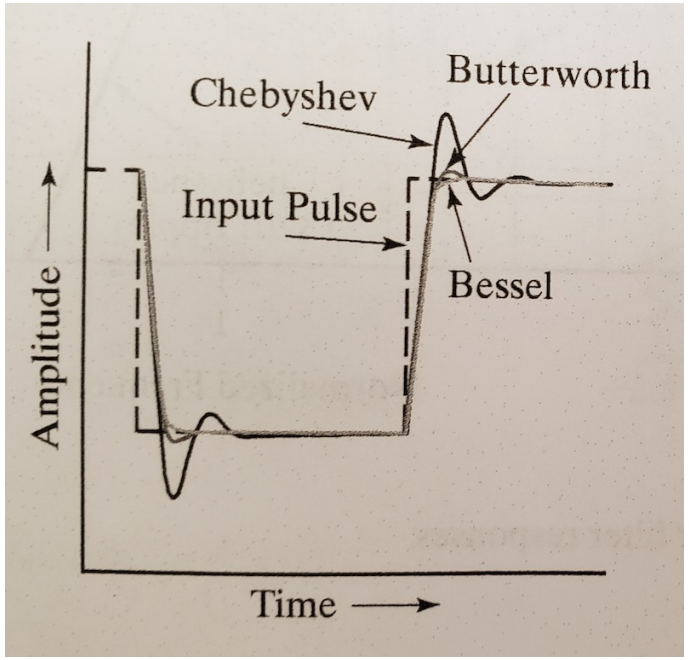


Popular Active Filters

- Butterworth filters
  - Maximally flat filters
- Bessel filters (best time domain behavior)
  - Near-constant time delay
  - Linear phase filter
- Chebyshev filters
  - Faster roll-off in the transition region



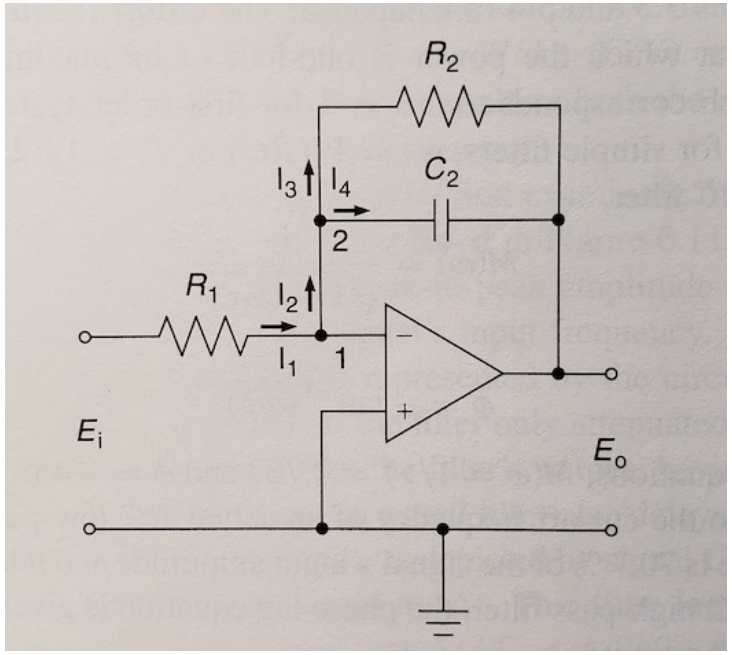
Frequency Domain



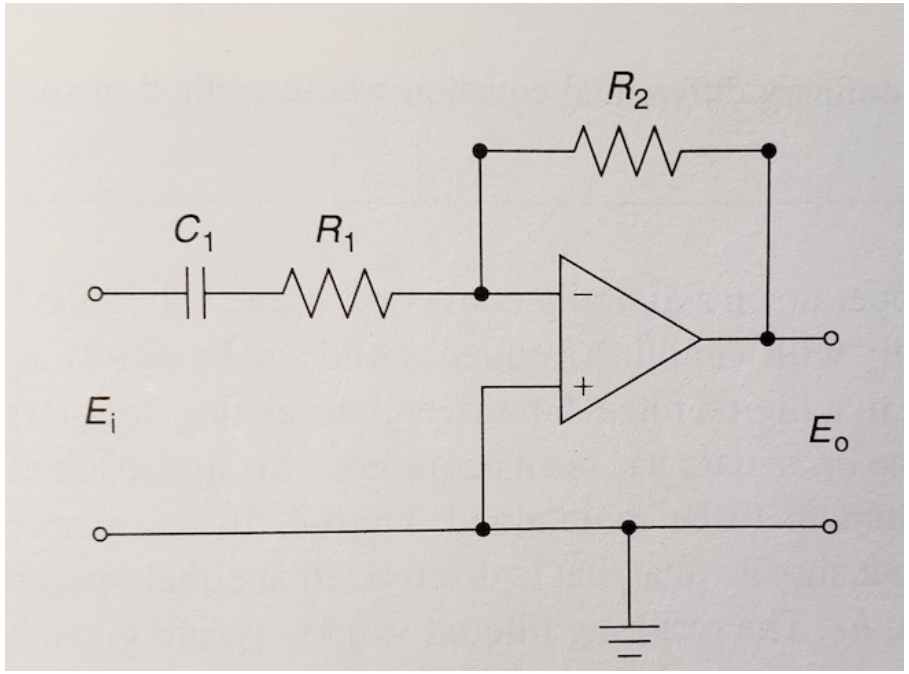
Time Domain

Butterworth Circuits - Examples

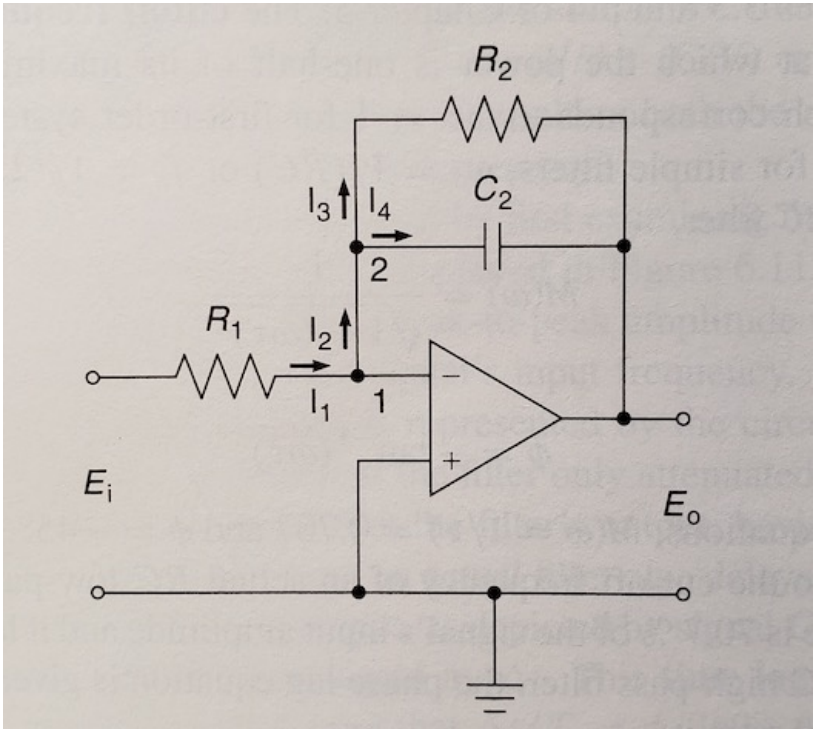
Low pass



High pass



Butterworth low pass filter
- find its differential equation and s-domain representation



- Example: If the cutoff frequency of a first-order low-pass filter is 100 Hz, what is the filter output at 500 Hz?

Reference

- Alexander, Charles K., Matthew NO Sadiku, and Matthew Sadiku. Fundamentals of electric circuits.
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- Patrick F Dunn, Measurement and Data Analysis for Engineering and Science.
- Ogata, Modern Control Engineering.
- David G. Alciatore, Introduction to Mechatronics and Measurement Systems
- Carryer, J. Edward, R. Matthew Ohline, and Thomas William Kenny. Introduction to mechatronic design
- De Silva, Clarence W., et al., editors. *Mechatronics : Fundamentals and Applications*. CRC Press, 2016, https://doi.org/10.1201/b19009.